

ARJUN GUPTA

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EDUCATION

Kanpur Institute of Technology, Kanpur: Bachelor of Computer Applications

Expected 2028

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Software Engineering, Web Development

RESEARCH EXPERIENCE

[Research Intern, NLP / RAG](#)

April 2026 – July 2026

IIT Bhubaneswar

- Implemented and benchmarked advanced RAG chunking strategies based on the KET-RAG paper.
- Reproduced and benchmarked DynaBERT-based transformer compression pipelines under fixed hardware and latency budgets.
- Width and depth-adaptive compression at 0.75× on RoBERTa-base yielded a 21.3% smaller model with a 1.78× inference speedup, while retaining 94.6% of baseline accuracy on QQP.

[Upcoming Internship: Machine Learning in FinTech](#)

November 2026 – December 2026

IIT Roorkee

- To start paid internship at IIT Roorkee in November 2026.

PROJECTS

[LeetTrack: DSA Preparation Platform](#)

August 2026

Technologies: FastAPI, Next.js, PostgreSQL, Google OAuth, Render

- Built and deployed a full-stack DSA assignment management platform adopted by 150+ students and 3+ instructors across my university.
- Implemented teacher-assigned LeetCode problem tracking, competitive leaderboards, and deadline management.
- Shipped OTP-based student authentication, a feedback system, public chat and notifications.
- Integrated the Nemotron 3 Ultra 550B LLM via the NVIDIA API to generate contextual hints.

[Jal Rakshak: Smart Water Tank Overflow Detection](#)

June 2026

Technologies: TensorFlow, YAMNet, Flutter, Android SDK

- Built a real-time water tank overflow detection system that classifies tank-fill sounds captured via a wireless mic, trained on a self-recorded custom dataset of 28 minutes of tank-fill audio collected at home.
- Deployed the model on Android and Windows with an auto-train feature that retrains on user-labeled data, allowing accuracy to improve continuously post-deployment.

[Industrial Anomaly Detection: PatchCore](#)

March 2026

Technologies: Python, PyTorch, WideResNet50

- Reproduced the PatchCore anomaly detection paper on the MVTec AD dataset, achieving 0.99 AUC using WideResNet50 feature extraction.
- Implemented coreset subsampling and KNN-based anomaly scoring from scratch, reducing the memory bank size by 90%+ with negligible loss in detection accuracy.

[Smart Attendance System](#)

October 2025

Technologies: TensorFlow, MobileNetV2, Flutter, Android SDK

- Optimized MobileNetV2 for low-compute mobile inference on a highly constrained 50-sample custom dataset
- Designed and deployed a mobile pipeline with Flutter and Android SDK, for fast local inference.

OPEN SOURCE CONTRIBUTIONS

- [Merged Pull Request](#): ml-recipes-py: Contributed a standardized ML pipeline featuring structured EDA, automated feature preprocessing, and Logistic Regression modeling to establish baseline code patterns and documentation standards.

TECHNICAL SKILLS

Languages: Python, Javascript, C

Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, HuggingFace, NumPy, Pandas

Web & Deployment: FastAPI, Next.js, HTML/CSS, Render, Vercel, PostgreSQL

Concepts: Retrieval-Augmented Generation (RAG), Transfer Learning, CNNs, Anomaly Detection, Transformers, Natural Language Processing (NLP), Data Structures and Algorithms [DSA in Python]

Tools: Jupyter, Git, GitHub, VS Code

NOMINATIONS

Nominee, JST Sakura Science Exchange Program | Iwate Prefectural University, Japan

- Nominated by host university for a JST-funded research exchange in advanced computing
- Selection results pending Nov 2026.
- Potential fully-funded visit to Japan in January 2027.

CERTIFICATIONS

- [Machine Learning Specialization](#): Stanford University
- [Deep Learning for Computer Vision and NLP](#): IIT Guwahati